ModelIt!

Teacher's Guide

CRISPR Gene Drives: Rewriting Evolution

Modeling How Engineered Genes Spread Through Wild Populations

Quick Reference

Grade Level	Grade 09L2
NGSS Standard	HS-LS3-1, HS-LS4-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS3-1, HS-LS4-5

Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits. Evaluate evidence supporting claims that changes in environmental conditions may result in increases in beneficial alleles and decreases in harmful alleles.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students evaluate competing claims about gene drive safety using model evidence and population genetics principles.
Disciplinary Core Idea	**LS3.A Inheritance of Traits** — Gene drives exploit the molecular mechanism of inheritance to bias allele transmission, overriding Mendelian ratios.
Crosscutting Concept	**Cause and Effect** — Students model how an engineered cause (biased inheritance) produces population-level effects that cascade through ecosystems.

CAST Testing Alignment

MULTIPLE CHOICE:

A gene drive with 95 percent inheritance bias and moderate fitness cost is released. What MOST determines whether it succeeds: (a) initial release size, (b) resistance mutation rate, (c) population mobility, (d) weather?

CONSTRUCTED RESPONSE:

Explain why a gene drive that eliminates 80 percent of malaria mosquitoes but then stalls due to resistance might be worse than one that eliminates 100 percent.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

CRISPR gene drives represent perhaps the most powerful and controversial biotechnology ever developed -- the ability to permanently modify an entire wild species. The potential to eliminate malaria is matched by the potential for catastrophic unintended consequences.

How Gene Drives Work

Normal inheritance gives each allele a 50 percent chance of being passed on. Gene drives use CRISPR to cut the chromosome without the drive and paste a copy of the drive into the break. The result: 95-99 percent of offspring inherit the drive instead of 50 percent.

The Malaria Application

Anopheles mosquitoes kill 600,000 people per year by transmitting Plasmodium parasites. A gene drive could either suppress mosquito populations or modify them to be unable to carry the parasite. Either approach could save hundreds of thousands of lives annually.

Resistance Evolution

Natural selection never stops. Any mutation that prevents the drive from copying itself will be strongly favored by selection. Lab experiments show resistance evolving within 10-25 generations. The question is whether the drive suppresses the population before resistance spreads.

Ecological Cascades

Mosquitoes are food for birds, bats, fish, and dragonflies. They pollinate some plants. Removing them could cascade through food webs in unpredictable ways. Or the ecological gap might be filled by other insects. We do not know -- and we cannot un-release a gene drive.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify gene drive frequency, wild-type allele, population size, fitness cost, resistance evolution, inheritance bias, and ecological impact as components.

E **Establish Relationships**
Students map the competition between drive spread (powered by inheritance bias) and natural selection (powered by fitness cost and resistance).

V **Visualize & Model**
Students build a model showing how gene drive dynamics play out across generations with competing evolutionary forces.

E **Evaluate Outcomes**
Students run ideal, realistic, and resistance scenarios to identify conditions for drive success or failure.

R **Revise & Extend**
Students add daisy chain mechanisms, off-target effects, or community consent to explore safety and ethics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"We can now edit an entire species. Should we?"*

Watch For

Show the dual nature: 600,000 malaria deaths vs. unknown ecological risk.

Slide 2: Learning Objectives

3 min

Say: *"Today we model the most powerful and controversial biotechnology ever invented."*

Watch For

Pre-teach gene drive, inheritance bias, and resistance.

Slide 3: Big Question

3 min

Say: *"Malaria kills a child every 2 minutes. We have the technology to stop it. But at what risk?"*

Watch For

Quick-write: If you could eliminate mosquitoes, would you? Why or why not?

Slide 4: LEVER Framework

2 min

Say: *"Seven components. Evolution vs. engineering. Who wins?"*

Watch For

Preview the race between drive spread and resistance.

Slide 5: Activity 1: Components

8 min

Say: *"What can scientists control? What does evolution control?"*

Watch For

Sort engineered vs. evolutionary components.

Slide 6: Activity 2: Connect

10 min

Say: *"The drive pushes the gene forward. Natural selection pushes back. Map the battle."*

Watch For

Map competing forces and identify the tipping point.

Slide 7: Activity 3: Simulate

10 min

Say: *"Does the drive win or does evolution win? It depends on the parameters."*

Watch For

Students run ideal, realistic, and resistance scenarios.

Slide 8: Discoveries

5 min

Say: *"The outcome is not guaranteed. And we cannot recall a released gene drive."*

Watch For

Discuss irreversibility and the precautionary principle.

Slide 9: STEM Challenge

12 min

Say: *"600,000 lives per year. Unknown ecological risk. Design the safest possible deployment."*

Watch For

Groups design phased deployment with safety mechanisms and halt criteria.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Fitness Cost, Inheritance Bias
- **INTERNAL:** Gene Drive Frequency, Wild-Type Allele Frequency, Population Size, Resistance Evolution, Ecological Impact

Relationships

- **Inheritance Bias to Gene Drive Frequency:** POSITIVE (+) — Higher inheritance bias spreads the drive faster through the population.
- **Gene Drive Frequency to Wild-Type Frequency:** NEGATIVE (-) — As the drive spreads, wild-type alleles are replaced.
- **Fitness Cost to Gene Drive Frequency:** NEGATIVE (-) — Higher fitness cost means drive carriers survive and reproduce less.
- **Gene Drive Frequency to Resistance Evolution:** POSITIVE (+) — As drive frequency increases, selection pressure for resistance mutations intensifies.
- **Resistance Evolution to Gene Drive Frequency:** NEGATIVE (-) — Resistance alleles block the drive mechanism, slowing its spread.
- **Gene Drive Frequency to Population Size:** NEGATIVE (-) — If the drive is suppressive, higher drive frequency reduces total population.
- **Population Size to Ecological Impact:** POSITIVE (+) — Population reduction cascades through food webs affecting predators, competitors, and prey.

Key Discoveries

- Gene drives exploit a mathematical advantage: 95-99 percent inheritance vs 50 percent overcomes even moderate fitness costs
- Resistance evolution is inevitable given enough time and population size -- the question is whether suppression occurs first
- Ecological impacts cascade unpredictably because removing one species affects all species connected to it in the food web
- There is a critical race between drive spread speed and resistance evolution speed that determines success or failure

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Gene drives are permanent and unstoppable"	Gene drives face evolutionary pressure from resistance mutations. Lab experiments show resistance evolving within 10-25 generations. The drive can stall or fail. However, even partial spread may cause irreversible ecological changes.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Removing mosquitoes would have no ecological effect"	Mosquitoes are food for birds, bats, dragonflies, and fish. They pollinate some plants. Their larvae filter water. Removing them could cascade through food webs. Or the gap could be filled by other species. We genuinely do not know.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"CRISPR gene drives are precise and controllable"	CRISPR is remarkably precise but not perfect. Off-target cuts occur, resistance evolves, and mechanisms can fail unpredictably. Lab experiments show gene drives reaching only 60-80 percent effectiveness before resistance plateaus the effect.